

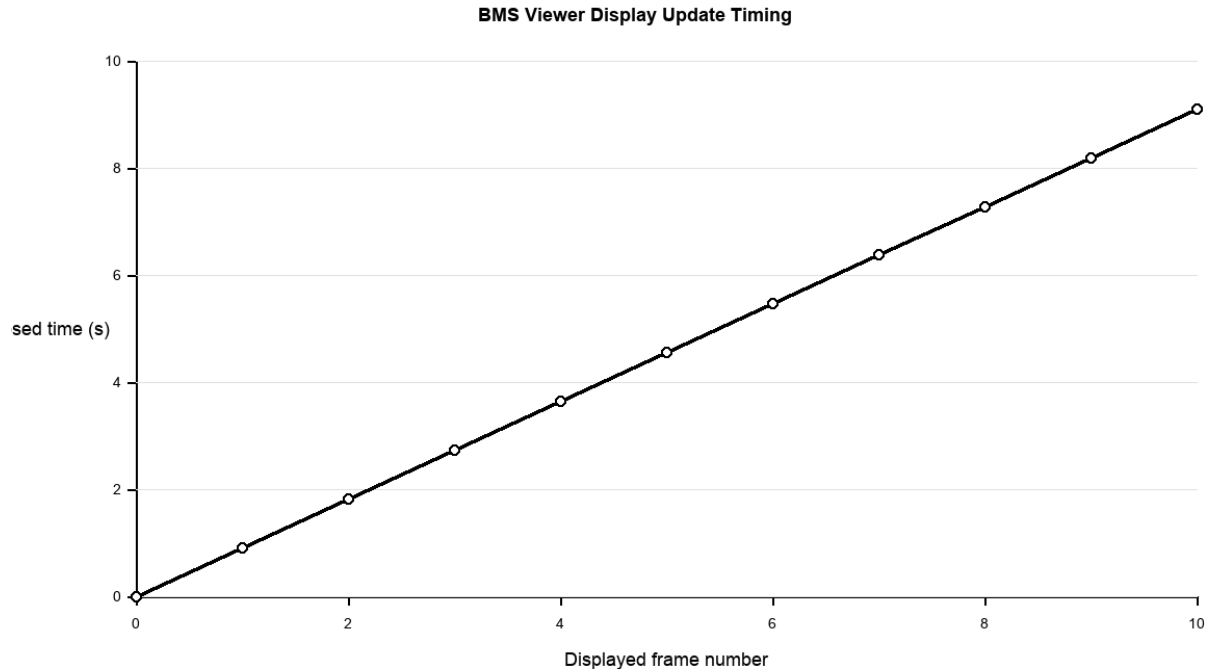
BMS Viewer Telemetry Graph Analysis

Battery Management System for Electric Scooter | ECE 445 Team 35 | Technical Design Note

Telemetry Update Rate

The BMS Viewer receives UART telemetry packets from the STM32, validates each frame, and updates the display with pack voltage, cell voltages, temperature readings, current, SOC, SOH, and fault flags. The measured viewer refresh rate was approximately 1.1 Hz.

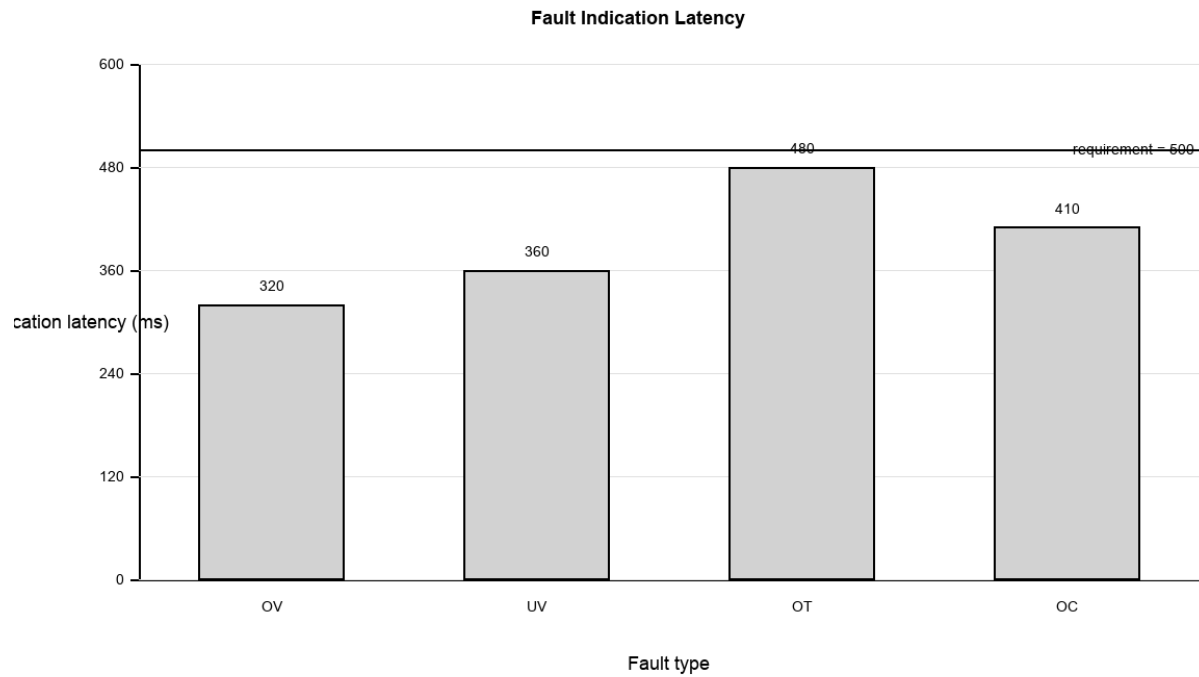
Update rate = displayed frames / elapsed time



The plotted frame timing corresponds to approximately one visible display refresh every 0.91 s, satisfying the requirement that the viewer update telemetry at least once per second.

Fault Display Latency

The viewer was also verified by injecting or simulating fault states and measuring how quickly each condition was reflected in the user interface. The graph below compares overvoltage, undervoltage, overtemperature, and overcurrent indication latency against the 500 ms display requirement.



Fault	Measured viewer indication	Requirement
OV	320 ms	≤ 500 ms
UV	360 ms	≤ 500 ms
OT	480 ms	≤ 500 ms
OC	410 ms	≤ 500 ms

Packet Validation

The plotted viewer results rely on accepting only valid UART frames. Packets are checked using start-of-frame bytes, payload length, and CRC so corrupted packets do not populate the dashboard.

Valid packet = correct SOF AND correct length AND valid CRC